AgriSense AI

Evidence-first market intelligence for mandi decisions.



The decision problem

A price quote alone does not tell the whole story.

Decision-makers need the date, mandi, unit, arrival context, and source coverage.

Missing days and mixed units can turn comparisons into misleading signals.

AgriSense starts by checking data readiness before presenting analysis.

A data-first workflow

01

Import AGMARKNET CSV observations and map source columns.

→

02

Audit date range, commodity/state/market coverage, duplicates, zero prices, arrivals, and units.

→

03

Preserve source commodity names and reject mixed-unit datasets rather than comparing incompatible prices.

→

04

Export the observations for review.

Market intelligence

Forecast: 7- or 14-day outlook, compared with a 14-day moving-average baseline.

Seasonality: monthly and Kharif/Rabi/Zaid price patterns.

Risk and crop comparison: recent movement, volatility, arrivals, and data coverage.

Results use imported observations—or clearly marked illustrative preview data.

Evidence and guardrails

26

overlapping weekly observations

Forecast evaluation compares a fitted linear trend with the baseline using chronological holdout MAE.

Intervals show uncertainty; they are not guaranteed prices.

Opportunity rankings expose their indicators and are not profit estimates.

Price-transmission analysis requires two real commodity series with at least 26 overlapping weekly observations; correlation is not causation.

Scenario tools

- raw price
- yield
- output price
- costs

Test raw price, yield, output price, and costs in a hypothetical processing scenario.

Results are gross-margin scenarios, not observed profitability or net profit.

Store sourced MSP and policy records with effective dates and manual verification.

Ask the built-in assistant about price, forecast, risk, seasonality, and data coverage in English, Hindi, or Kannada.

IBM LinuxONE deployment path

React/Vite web app → Node.js API → PostgreSQL with Drizzle.



linux/s390x

OCI container build files target linux/s390x.

OpenShift manifests define web/API services and routed paths.

Prepared for IBM LinuxONE deployment; not yet deployed to an IBM cluster.

Pilot hypothesis

Begin with one selected commodity, state, and mandi workflow.

Use a representative official AGMARKNET extract to test ingestion and coverage.

Validate the forecast against the moving-average baseline before making accuracy claims.

Identify the intended buyer, usage cadence, and commercial model through pilot discovery.

Validation roadmap

Data:

confirm column mapping, units, gaps, and source freshness.

Analytics:

measure forecast error and coverage on a held-out history.

Workflow:

test whether users can move from audit to comparison and scenario analysis.

Commercial:

agree the buyer, value metric, and pricing only after pilot feedback.

Pilot invitation

Bring representative market data and a real decision workflow. We'll validate signal quality before scaling.

Seeking a data partner and pilot users.

First proof points: data coverage, benchmarked forecast error, and workflow fit.